

Quantum Fault Tolerance Meets Toom's Contours

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1 Shrinking

We are going to show that the SWEEP decoder 'shrinks' the error by proving that the edges/poles of error clusters get closer as a result of a decoding step. For that, we consider the standard embedding of the grid in $\mathbb{R}^d$ and use the term of potential walls.

Definition 1.1 (The standart embedding). Consider the d -dimensional infinite grid $\mathcal{G}^d$. The standard embedding $\mathcal{G}^d \rightarrow \mathbb{R}^d$ is given by fixing a vertex in $\mathcal{G}^d$ to be sent to the origin and then mapping the generators of the grid to the standard basis. We extend notation as follows: for a vertex $v \in \mathcal{G}^d$, a generator $g \in S$, and a coordinate $i \in [d]$, such that under the embedding $g \rightarrow i$, we denote by $v_i = v_g$ the i 'th coordinate of the image of v .

Definition 1.2 (Potential Walls). We define the potential walls to be the $d + 1$ functions $L_1, L_2, \dots, L_{d+1}$, from the vertices of $\mathcal{G}_n^d$ to the reals as follow:

1. For any coordinate $i \in [d]$, we define $L_i: \mathcal{G}^d \rightarrow \mathbb{R}$ as $L_i(v) := v_i$. In addition, for the generator $g \in S$ which is mapped to i , i.e $g \rightarrow i$, we identify $L_g = L_i$.
2. We define $L_{d+1}: \mathcal{G}^d \rightarrow \mathbb{R}$ by $L_{d+1}(v) = \sum_{i=1}^d v_i$.

For a subset of vertices $\Gamma \subset \mathcal{G}_n^d$, a vertex $v \in \Gamma$, and a potential wall $l: \mathcal{G}_n^d \rightarrow \mathbb{R}$, we say that v is a *local maximum* of l in Γ if for any neighbor u of v , such that $u \in \Gamma$, we have $l(v) \geq l(u)$. We say that v is a *strong local maximum* if the inequality is strict, namely $l(v) > l(u)$. When the subset Γ is clear from the context, we might omit it and briefly say that v is a local maximum of l .

Claim 1.3. Let z be an assignment, and let ∂z be the set of vertices adjacent to unsatisfied checks induced by z . Let v be a vertex in ∂z , and let g be a coordinate in S such that v is a local maximum of L_g . Then $gv \notin \partial z$.

Proof. Suppose $gv \in \partial z$, then

$$L_g(gv) = (gv)_g = 1 + v_g = 1 + L_g(v)$$

In contradiction to v being a local maximum of L_g in ∂z . □

Claim 1.4. Let z be an assignment, and let ∂z be the set of vertices adjacent to unsatisfied checks induced by z . Let v be a vertex in ∂z , and let g be a coordinate in S such that v is a local maximum of L_g . Then for any check associated with a $d' - 1$ face rooted at v that contains gv , it is satisfied.

Proof. Assume, through contradiction, the existence of an unsatisfied check associated with a $d' - 1$ positive face rooted at v , which contains gv . This implies $gv \in \partial z$, contradicting Claim 1.3. $\square$

Definition 1.5 (Small Code at Vertex v). For a vertex v , we define the X -small code at vertex v to be the binary assignments over the bits (d' faces) rooted at v that satisfy $\mathcal{C}_X$'s restrictions rooted at v ($d' - 1$ faces). Similarly, we define the Z -small code at vertex v to be the binary assignments over the bits (d' faces) rooted at v that satisfy $\mathcal{C}_Z$'s restrictions rooted at v ($d' + 1$ faces). We denote by $\mathcal{C}_v^-$ and $\mathcal{C}_v^+$ the X -small code at v and Z -small code at v .

Definition 1.6. Should rewrite it better. We define Stab_v^- and Stab_v^+ as the following linear codes:

$$\text{Stab}_v^- := \left\{ z : \partial z = 0 \text{ and } z \in \text{Span} \left(\{ \theta_{\mathbf{v}, \mathbf{S}'/\mathbf{g}}, \theta_{\mathbf{g}\mathbf{v}, \mathbf{S}'/\mathbf{g}} \text{ for } S' \subset S, |S'| = d' + 1, g \in S' \} \right) \right\}$$

$$\text{Stab}_v^+ := \left\{ z : \partial^* z = 0 \text{ and } z \in \text{Span} \left(\{ \theta_{\mathbf{v}, \mathbf{S}'/\mathbf{g}}, \theta_{\mathbf{g}\mathbf{v}, \mathbf{S}'/\mathbf{g}} \text{ for } S' \subset S, |S'| = d' + 1, g \in S' \} \right) \right\}$$

[COMMENT] Regarding the beneath, The number of restriction in ∂^+ case was counted in negligently. In addition, we should show that for the ∂^- case, restrictions of the form $\theta_{\mathbf{g}_2\mathbf{g}_1\mathbf{v}, \mathbf{S}}$ are degenerate. In the bottom line it's look like:

$$\partial^+ \theta_{\mathbf{g}_2\mathbf{g}_1\mathbf{v}, \mathbf{S}} = \partial^+ \theta_{\mathbf{g}_1\mathbf{v}, (\mathbf{S}/\mathbf{g}_3) \cup \mathbf{g}_2} + \partial^+ \theta_{\mathbf{g}_2\mathbf{v}, (\mathbf{S}/\mathbf{g}_3) \cup \mathbf{g}_1}$$

Lets develop it, consider a restriction rooted at g_2g_1v , and let's look on it's boundary:

$$\partial^+ \theta_{\mathbf{g}_2\mathbf{g}_1\mathbf{v}, \mathbf{S}'} = \sum_{g \in S/S'} \theta_{\mathbf{g}_2\mathbf{g}_1\mathbf{v}, \mathbf{S}' \cup \mathbf{g}} + \sum_{g \in S/S'} \theta_{\mathbf{g}^{-1}\mathbf{g}_2\mathbf{g}_1\mathbf{v}, \mathbf{S}' \cup \mathbf{g}}$$

Now, any term in the left closure is a bit beyond the scope of the bits in Stab_v^- , thus, we are zeroing them out. The bits, on the right closers are in the scope, only if $g \in \{g_1, g_2\}$. Thus, in total the check is equivalence to:

$$\begin{aligned} \partial^+ \theta_{\mathbf{g}_2\mathbf{g}_1\mathbf{v}, \mathbf{S}'} &= \theta_{\mathbf{g}_1\mathbf{v}, \mathbf{S}' \cup \mathbf{g}_2} \cdot \mathbf{1}_{g_2 \notin S'} \\ &\quad + \theta_{\mathbf{g}_2\mathbf{v}, \mathbf{S}' \cup \mathbf{g}_1} \cdot \mathbf{1}_{g_1 \notin S'} \\ \partial^+ \theta_{\mathbf{g}_1\mathbf{v}, (\mathbf{S}'/\mathbf{g}_3) \cup \mathbf{g}_2} &= \sum_{g \in S/S'} \theta_{\mathbf{g}_1\mathbf{v}, (\mathbf{S}'/\mathbf{g}_3) \cup \mathbf{g}_2 \cup \mathbf{g}} + \sum_{g \in S/S'} \theta_{\mathbf{g}^{-1}\mathbf{g}_1\mathbf{v}, (\mathbf{S}'/\mathbf{g}_3) \cup \mathbf{g}_2 \cup \mathbf{g}} \\ \partial^+ \theta_{\mathbf{g}_2\mathbf{v}, (\mathbf{S}/\mathbf{g}_3) \cup \mathbf{g}_1} &= \sum_{g \in S/S'} \theta_{\mathbf{g}_2\mathbf{v}, (\mathbf{S}'/\mathbf{g}_3) \cup \mathbf{g}_1 \cup \mathbf{g}} + \sum_{g \in S/S'} \theta_{\mathbf{g}^{-1}\mathbf{g}_2\mathbf{v}, (\mathbf{S}'/\mathbf{g}_3) \cup \mathbf{g}_1 \cup \mathbf{g}} \end{aligned}$$

$$\begin{aligned}\partial^+ \theta_{\mathbf{g}_1 \mathbf{v}, (\mathbf{S}'/\mathbf{g}_3) \cup \mathbf{g}_2} &= \left(\sum_{g \in S/S'} \theta_{\mathbf{g}_1 \mathbf{v}, (\mathbf{S}'/\mathbf{g}_3) \cup \mathbf{g}_2 \cup g} \right) + \theta_{\mathbf{v}, (\mathbf{S}'/\mathbf{g}_3) \cup \{\mathbf{g}_2, \mathbf{g}_1\}} \\ \partial^+ \theta_{\mathbf{g}_2 \mathbf{v}, (\mathbf{S}/\mathbf{g}_3) \cup \mathbf{g}_1} &= \left(\sum_{g \in S/S'} \theta_{\mathbf{g}_2 \mathbf{v}, (\mathbf{S}'/\mathbf{g}_3) \cup \mathbf{g}_1 \cup g} \right) + \theta_{\mathbf{v}, (\mathbf{S}'/\mathbf{g}_3) \cup \{\mathbf{g}_1, \mathbf{g}_2\}}\end{aligned}$$

Other way:

$$\begin{aligned}\partial^+ \sum_{g' \in S/S'} \theta_{\mathbf{g}' \mathbf{v}, \mathbf{S}'} &= \sum_{g' \neq g \in S/S'} \theta_{\mathbf{g}' \mathbf{v}, \mathbf{S}' \cup g} + \sum_{g \in S/S'} \theta_{\mathbf{v}, \mathbf{S}' \cup g} \\ &= \sum_{g' \neq g \in S/S'} \theta_{\mathbf{g}' \mathbf{v}, \mathbf{S} \cup g} + \partial^+ \theta_{\mathbf{v}, \mathbf{S}'}\end{aligned}$$

Where we use the fact that if $g = g'$ then the bit is beyond the scope. In addition:

$$\begin{aligned}\partial^+ \sum_{g' < g \in S/S'} \theta_{\mathbf{g}' \mathbf{g} \mathbf{v}, \mathbf{S}'} &= \sum_{g' < g \in S/S'} \theta_{\mathbf{g}' \mathbf{v}, \mathbf{S}' \cup g} + \theta_{\mathbf{g} \mathbf{v}, \mathbf{S}' \cup g'} \\ &= \sum_{g' \neq g \in S/S'} \theta_{\mathbf{g}' \mathbf{v}, \mathbf{S}' \cup g} \\ &= \partial^+ \sum_{g' \in S/S'} \theta_{\mathbf{g}' \mathbf{v}, \mathbf{S}'} + \partial^+ \theta_{\mathbf{v}, \mathbf{S}'}\end{aligned}$$

Thus, we find more $\binom{d}{d'-1}$ restriction over the checks. [We need to show that those restrictions are independent]. In addition:

$$\partial^+ \sum_{S'} \sum_{g' < g \in S/S'} \theta_{\mathbf{g}' \mathbf{g} \mathbf{v}, \mathbf{S}'} = 0$$

$$\partial^+ \theta_{\mathbf{g}_2 \mathbf{g}_1 \mathbf{v}, \mathbf{S}} = \partial^+ \theta_{\mathbf{g}_1 \mathbf{v}, (\mathbf{S}/\mathbf{g}_3) \cup \mathbf{g}_2} + \partial^+ \theta_{\mathbf{g}_2 \mathbf{v}, (\mathbf{S}/\mathbf{g}_3) \cup \mathbf{g}_1}$$

I don't succes to advance the idea more than that, So in the end I believe that I will do is to say that the diminsion of Stab_v^- is at least $\binom{d}{d'+1}$. And by fixing coordinates we add more $\binom{d}{d'}$ restriction. So the dimension becomes:

$$\begin{aligned}&\binom{d}{d'+1} - \binom{d}{d'} \\ &\binom{d}{d'+1} - \binom{d}{d'} + \binom{d}{d'-1}\end{aligned}$$

Note that since we talk on mapping checks $\rightarrow$ bits, we use the ∂^+ . Eventually, we need to use the poincare duality $H^{d'} \rightarrow H_{d-d'}$, the duality also should be used to define the SWEEP rule to C_Z . I didn't successes to do it other way.

Claim 1.7. Let v be a vertex. For any $z \in \mathcal{C}_v^\pm$, there is a stabilizer $w \in \text{Stab}_v^\pm$ such that $z = w|_v$; namely, z and w agree on the faces rooted at v .

Proof. We will show that codes obtained from $\text{Stab}_v^\pm$ by forcing the faces rooted at v to $z|_v$ have non-zero dimensions. Then it follows that any code word of $\mathcal{C}_v^\pm$ could be extend to codewords in $\text{Stab}_v^\pm$. The number of bits in both $\text{Stab}_v^\pm$ is:

$$\binom{d}{d'} + d \binom{d-1}{d'}$$

The numbers of checks for $\text{Stab}_\pm$ are:

$$\binom{d}{d' \pm 1} + d \binom{d-1}{d' \pm 1}$$

The restriction to $z|_v$ sets the bits rooted at v and satisfies the checks that are rooted at v , so the checks of the form $\theta_{\mathbf{v}, S'}$ for $|S'| = d' \pm 1$ are degenerate, and the dimension that remains is:

$$\dim \geq d \binom{d-1}{d'} - d \binom{d-1}{d' \pm 1}$$

In the case that $d' = (d-1)/2$, then for any $x \in [d-1]$ we have $\binom{d-1}{d'} > \binom{d-1}{x}$ therefore:

$$\begin{aligned} \dim &\geq d \binom{d-1}{d/2} - d \binom{d-1}{d/2 \pm 1} \\ &> d \end{aligned}$$

So there is at least 2^d stabilizers $w \in \text{Stab}_v$ that agree with z . □

Claim 1.8. Let z be a finite assignment. Suppose that

$$\max \{L_{d+1}(u) : u \in z\} > \max \{L_{d+1}(u) : u \in \partial^\pm z\}$$

Then there is $\tilde{z} \sim_{\mathcal{C}_X} z$ such that:

$$\max \{L_{d+1}(u) : u \in \tilde{z}\} \leq \max \{L_{d+1}(u) : u \in z\} - 1$$

Proof. Let $v_1, v_2, \dots, v_k$ be the vertices in z that achieve the maximum for L_{d+1} , namely $v_1, v_2, \dots, v_k = \arg \max_{v \in z} L_{d+1}(v)$, and assume that $v_i \notin \partial^\pm z$ for any $i \in [k]$.

Consider the i th vertex v_i . By definition, since $v_i \notin \partial^\pm z$, we have $\partial^\pm z|_{v_i} = 0$, and since v_i is a maximum point of L_{d+1} in z , it is also a local maximum, and therefore $z|_{v_i+} = z|_{v_i}$. Combining the two, we get that $z|_{v_i}$ is a codeword of the small code at v_i , namely $z|_{v_i} \in \mathcal{C}_{v_i}^\pm$. Thus, by Claim 1.7, there is a stabilizer $w_i \in \text{Stab}_{v_i}^\pm$ such that w_i and z agree on the positive faces of v_i , namely $w_i|_{v_i} = z|_{v_i}$.

Set $\tilde{z} \leftarrow z + \sum_i w_i$. Since $\sum_i w_i$ is a stabilizer, we have $\tilde{z} \sim_{\mathcal{C}_X} z$. In addition, for any $j \neq i$, it follows that $v_j \notin w_i$. Otherwise, there exists $S' \subset S$ such that $v_j \in \theta_{v_i, S'}$, and therefore $L_{d+1}(v_j) > L_{d+1}(v_i)$, in contradiction to v_i and v_j being both maximum points for L_{d+1} in $\partial^\pm z$. Thus, we have:

$$\tilde{z}|_{v_i} = (z + \sum_i w_i)|_{v_i} = z|_{v_i} + (z + w_i)|_{v_i} = 0$$

Namely, for any i , the i th vertex v_i is not in $\tilde{z}$. Hence, the vertex domain of $\tilde{z}$ is contained in the union of the vertices of z with the vertices of $w_1, w_2, \dots, w_k$, substituting $v_1, v_2, \dots, v_k$. Overall, we get:

$$\begin{aligned} \max \{L_{d+1}(u) : u \in \tilde{z}\} &\leq \max \{L_{d+1}(u) : u \in z \bigcup_i w_i / \bigcup_i v_i\} \\ &\leq \max \{L_{d+1}(u) : u \in z\} - 1 \end{aligned}$$

□

Claim 1.9. Let z be a finite assignment. Then there is $\tilde{z} \sim_{C_X} z$ such that

$$\max \{L_{d+1}(u) : u \in \tilde{z}\} = \max \{L_{d+1}(u) : u \in \partial^\pm z\}$$

Proof. Let z be a finite assignment, and let k be an integer such that $k \geq 1$ and

$$\max \{L_{d+1}(u) : u \in z\} = \max \{L_{d+1}(u) : u \in \partial^\pm z\} + k$$

For an integer $l \leq k$, consider the assignments $\tilde{z}_0, \tilde{z}_1, \dots, \tilde{z}_l$ defined as follow:

- The first element, $\tilde{z}_0 = z$.
- If the $i - 1$ th element satisfies

$$(\star) \quad \max \{L_{d+1}(u) : u \in \tilde{z}_{i-1}\} > \max \{L_{d+1}(u) : u \in \partial^\pm \tilde{z}_{i-1}\}$$

Then define the i th element, $\tilde{z}_i$, to be the result of applying the construction, given in Claim 1.8, on $\tilde{z}_{i-1}$, namely $\partial \tilde{z}_{i+1} = \partial \tilde{z}_i$ and

$$\max \{L_{d+1}(u) : u \in \tilde{z}_i\} \leq \max \{L_{d+1}(u) : u \in \tilde{z}_{i-1}\} - 1$$

If the $i - 1$ th element doesn't satisfies $\star$, then set $l = i - 1$.

- If the k th element was defined, set $l = k$.

Since $\sim_{C_X}$ is a transitive relation, we have that for any $i \in [l]$, $z \sim_{C_X} \tilde{z}_i$. If $l \neq k$, then there exists $\tilde{z}_i$ such that $\tilde{z}_i \sim_{C_X} z$ and $\tilde{z}_i$ satisfies $\star$, namely by setting $\tilde{z} = \tilde{z}_i$, we get the desired. So, it is left to handle the case where $l = k$. In that case, set $\tilde{z} = \tilde{z}_k$. By Claim 1.8:

$$\max \{L_{d+1}(u) : u \in \tilde{z}\} \leq \max \{L_{d+1}(u) : u \in \partial^\pm z\}$$

Yet, since $\tilde{z} \sim_{C_X} z$, it follows that $\partial \tilde{z} = \partial z$, So:

$$\max \{L_{d+1}(u) : u \in \partial^\pm z\} = \max \{L_{d+1}(u) : u \in \partial^\pm \tilde{z}\} \leq \max \{L_{d+1}(u) : u \in z\}$$

Therefore, we have the equality:

$$\max \{L_{d+1}(u) : u \in \tilde{z}\} = \max \{L_{d+1}(u) : u \in \partial^\pm z\}$$

□

Claim 1.10. Let z be a finite assignment and let ξ be the result of the application of the Sweep rule on it. Then:

1. For any $i \in [d]$: $\max \{L_i(v) : v \in \partial z\} = \max \{L_i(v) : v \in \partial \xi\}$
2. And for $i = d + 1$: $\max \{L_{d+1}(v) : v \in \partial z\} > \max \{L_{d+1}(v) : v \in \partial \xi\} + 1$

Proof. First, since the Sweep rule is invariant to addition by stabilizers, for any $\tilde{z}$ such that $\tilde{z} \sim_{C_X} z$, the correction applied by the Sweep rule on $\tilde{z}$ is the same as the application on z . Denote by ϕ the correction, by $\tilde{\xi}$ the result of applying the Sweep rule on $\tilde{z}$, and by w the stabilizer which is the difference between z and $\tilde{z}$, namely:

$$\begin{aligned}\xi &= z + \phi \\ \tilde{\xi} &= \tilde{z} + \phi \\ \tilde{z} &= z + w\end{aligned}$$

The $\partial^\pm$ boundary of $\tilde{\xi}$ equals:

$$\partial \tilde{\xi} = \partial (\tilde{z} + \phi) = \partial (z + \phi) = \partial \xi$$

So, in total, we have that for any $\tilde{z}$ such that $\tilde{z} \sim_{C_X} z$, it holds that $\partial \tilde{z} = \partial z$ and $\partial \tilde{\xi} = \partial \xi$. Therefore, showing that there exists $\tilde{z} \sim_{C_X} z$ for which the claim holds proves the claim for z . By Claim 1.9, there is $\tilde{z} \sim_{C_X} z$ such that:

$$\max \{L_{d+1}(u) : u \in \tilde{z}\} = \max \{L_{d+1}(u) : u \in \partial^\pm \tilde{z}\}$$

For such $\tilde{z}$, any maximum of L_{d+1} in $\partial \tilde{z}$, is also a maximum in $\tilde{z}$. Denote by $v_1, \dots, v_k$ the maximum points in $\partial \tilde{z}$. Since they are maximum points in $\tilde{z}$, it holds that for each $i \in [k]$, $\tilde{z}|_{v_i} = \tilde{z}|_{v_i+}$, and hence $\partial \tilde{z}|_{v_i} = \partial \tilde{z}|_{v_i+}$. Since $\tilde{z}|_{v_i+}$ is an assignment rooted at v_i , then v_i can suggest $\tilde{z}|_{v_i+}$, namely the condition in Sweep rule is satisfied for vertex v_i . Thus:

$$\partial \tilde{\xi}|_{v_i} = \partial \tilde{z}|_{v_i} + \partial \phi|_{v_i} = 0$$

Namely, for any i , the i th vertex v_i is not in $\tilde{\xi}$. For any vertex v denote by ϕ_v the correction made by the vertex v , namely $\phi = \sum_{v \in \partial \tilde{z}} \phi_v$. Hence, the vertex domain of $\tilde{\xi}$ is contained in the union of the vertices of $\tilde{z}$ with the vertices of $\cup_{v \in V} \phi_v$, substituting maximum points $v_1, v_2, \dots, v_k$. Overall, we get:

$$\begin{aligned}\max \{L_{d+1}(u) : u \in \partial \tilde{\xi}\} &\leq \max \{L_{d+1}(u) : u \in \partial \tilde{z} \cup \bigcup_{v \in \partial \tilde{z}} \phi_v / \bigcup_i v_i\} \\ &\leq \max \{L_{d+1}(u) : u \in \partial \tilde{z}\} - 1\end{aligned}$$

Let v be a vertex that is a maximum of L_g in $\partial \tilde{z}$, maximum points of L_g do not see syndrome at the g direction. If and:

$$\phi_v = \sum \theta_{\mathbf{v}, \mathbf{S}'_i}$$

By Claim 1.4 $\partial\tilde{z}|_{gv} = 0$, Hence:

$$\begin{aligned}\partial\phi_v|_{gv} &= \left(\partial \sum \theta_{\mathbf{v}, \mathbf{S}'_i}\right)|_{gv} \\ &= \left(\sum_{S'_i} \sum_{g' \in S'_i} \theta_{\mathbf{v}, \mathbf{S}'_i/g'} + \theta_{\mathbf{g}'\mathbf{v}, \mathbf{S}'_i/g'}\right)|_{gv} \\ &= 0\end{aligned}$$

Thus if $g \in \cup S'_i$, Then:

$$\theta_{\mathbf{g}\mathbf{v}, \mathbf{S}'_i/g} \in \partial\phi_v|_{gv}$$

So for $\theta_{\mathbf{g}\mathbf{v}, \mathbf{S}'_i/g} = 0$, it must appear at least twice in the sum, yet the left terms in the sum cannot contain it (they differ by the root vertex), and if there is a right term $\theta_{\mathbf{g}\mathbf{v}, \mathbf{S}'_j/g} = \theta_{\mathbf{g}\mathbf{v}, \mathbf{S}'_i/g}$, then it implies that $S_j = S_i$. Which is a contradiction, therefore: $g \notin \cup S'_i$ and $gv \notin \phi_v$.

$$\begin{aligned}\max \left\{ L_g(u) : u \in \partial\tilde{\xi} \right\} &\leq \max \left\{ L_g(u) : u \in \partial\tilde{z} \bigcup_{v \in \partial\tilde{z}} \phi_v / \bigcup_i v_i \right\} \\ &\leq \max \left\{ L_g(u) : u \in \partial\tilde{z} \right\} + 1 - 1\end{aligned}$$

□